

Intra-abdominal Sepsis: Cholangitis



Section I: Scenario Demographics

Scenario Title:	Cholangitis
Date of Development:	08/09/2014
Target Learning Group:	☒ Juniors (PGY 1 - 2) ☐ Seniors (PGY ≥ 3) ☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Martin Kuuskne
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To manage an undifferentiated septic patient.
CRM Objectives:	1) Effectively establish oneself as the team leader and exercise non-verbal leadership techniques 2) To initiate investigations and management in line with the surviving sepsis campaign 3-hour bundle.
Medical Objectives:	1) Recognize the presentation of a septic patient and ensure appropriate investigations and early treatment. 2) Initiate vasopressor therapy in the setting of septic shock.

Case Summary: Brief Summary of Case Progression and Major Events

A 60-year-old male presents with a four-day history of abdominal pain secondary to cholangitis. The patient presents in septic shock requiring intravenous fluid resuscitation, empiric broad-spectrum antibiotics and vasopressor support and suffers a PEA arrest prior to disposition to advanced imaging or definitive management.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby

Surviving Sepsis Campaign: International Guidelines for Management of Severe Sepsis and Septic Shock: 2012. Available at : <http://www.sccm.org/Documents/SSC-Guidelines.pdf>



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Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case			
You are working a day shift at a community hospital emergency department. You are handed a chart of a patient presenting with abdominal pain. You recognize the following vital signs: Heart rate 120, blood pressure 85/55, respiratory rate 20, and O ₂ Saturation 95%.			
B. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin ☐ Mannequin ☐ Standardized Patient ☐ Hybrid ☐ Task Trainer	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual ☐ Physical ☐ Emotional/Experiential ☐ Other: ☐ N/A
Confederates	Brief Description of Role		
N/A	N/A		
C. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
D. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☒ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☒ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
E. Moulage			
N/A			
F. Approximate Timing			
Set-Up:	5 min	Scenario:	12 min
		Debriefing:	15 min

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Section V: Patient Data and Baseline State

A. Patient Profile and History					
Patient Name: Glen McDonald		Age: 60		Weight: 100kg	
Gender: ☒ M ☐ F		Code Status: Full code.			
Chief Complaint: Abdominal Pain.					
History of Presenting Illness: 4 day history of abdominal pain; began in the epigastric area and has since migrated to the RUQ. It is a sharp, constant pain. The patient had previous similar episodes that resolved spontaneously. Fever for 2 days brought him to the emergency department.					
Past Medical History:	HTN		Medications:	Lisinopril 20mg PO Daily	
	Hyperlipidemia			Atorvastatin 20mg PO QHS	
	T2DM			Metformin 500mg PO BID	
Allergies: NKDA					
Social History: Smoker, 20 PY history					
Family History: Non-contributory					
Review of Systems: (from daughter)	CNS:	Normal			
	HEENT:	Normal			
	CVS:	Feels heart racing			
	RESP:	No cough			
	GI:	Nausea and vomiting x 3 this morning			
	GU:	Normal			
	MSK:	Normal	INT:	Normal	
B. Baseline Simulator State and Physical Exam					
☐ No Monitor Display		☒ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 120/min	BP: 85/55	RR: 20/min	O ₂ SAT: 95%		
Rhythm: Sinus	T: 39.5°C	Glucose: 9.0 mmol/L	GCS: 14 (E3 V5 M6)		
General Status: In pain					
CNS:	Pupils 2+ reactive bilaterally, moving all 4 limbs to pain, no FND, normal reflexes				
HEENT:	Normal				
CVS:	Tachycardia, no EHS				
RESP:	GAEB				
ABDO:	Tender in RUQ with rebound tenderness and guarding.				
GU:	Normal				
MSK:	Normal	SKIN:	Cool, clammy extremities, jaundiced		

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus HR: 120/min BP: 85/55 RR: 20/min O ₂ SAT: 95% T: 39.5°C	In pain, GCS: 14	<u>Learner Actions</u> ☐ Monitor/Full vitals ☐ Hx/PE ☐ 2 Large bore IVs ☐ 1L IV fluid bolus ☐ Lab investigations including LFTs, VBG, lactate). ☐ Blood cultures x 2 ☐ CXR/ EKG ☐ Analgesia ☐ Empiric IV Antibiotics	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - IV Fluid Bolus → HR to 115 BP to 90/60 <u>Triggers</u> <i>For progression to next state</i> - 5 minutes → 2. Deterioration
2. Deterioration → HR to 140 over 20 seconds → BP to 75/40	Obtunded, GCS: 8 (E2V2M4)	<u>Learner Actions</u> ☐ 1L IV fluid bolus #2 ☐ Initiation of vasopressors after 2 nd fluid bolus ☐ ± Central Line insertion ☐ Bedside ultrasound	<u>Modifiers</u> Bolus → BP to 80/40 <u>Triggers</u> - 8 Minutes → 3. PEA
3. PEA → BP: 0/0 → O ₂ SAT: 0 (Rate and rhythm continue)	No pulses felt	<u>Learner Actions</u> ☐ ACLS PEA protocol ☐ Timely recognition of PEA ☐ CPR with minimal interruptions ☐ Epinephrine 1mg IV q3min ☐ Consideration of Hs/Ts ☐ Crash Intubation ☐ 1L IV fluid bolus #3	<u>Modifiers</u> <u>Triggers</u> 1 round of CPR/EPI and 3 rd litre of NS → 4. Post resuscitation 2 Rounds of CPR/EPI given → 4. Post resuscitation
4. Post Resuscitation → HR: 120/min → BP: 80/50 → RR: 12 or set by vent → O ₂ SAT: 95%	Obtunded, GCS: 8 (E2V2M4) if not intubated. If intubated = sedated	<u>Learner Actions</u> ☐ Consultation (General Surgery, GI, ICU) ☐ Fluid Bolus ☐ Addresses vasopressor infusion if previously started ☐ Arrange for advanced imaging.	End Scenario PRN.

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na: 138	K: 4.9	Cl: 99	HCO ₃ : 14	BUN: 10	Cr: 230	Glu: 12
VBG	pH: 7.12	PCO ₂ : 30	PO ₂ : 60	HCO ₃ : 14	Lactate:6.4	
WBC: 20		Hg: 120		Hct: 40%		Plt: 220
ALT: 200	AST: 123	ALP: 560	GGT: 765	Bili(t): 113		Bili(d): 103

Images (ECGs, CXRs, etc.)	
Normal CXR - https://emsimcases.files.wordpress.com/2015/04/normal-cxr-male.jpg	EKG – Sinus Tachycardia http://cdn.lifeinthefastlane.com/wp-content/uploads/2011/12/sinus-tachycardia.jpg
Ultrasound Video Files (if applicable)	
U/S Showing GB Stones and dilated biliary system.	

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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CRM Objectives:	1) Effectively establish oneself as the team leader and exercise non-verbal leadership techniques 2) To initiate investigations and management in line with the surviving sepsis campaign.
Medical Objectives:	1) Recognize the presentation of a septic patient and ensure appropriate investigations and early treatment. 2) Initiate vasopressor therapy in the post arrest patient.
Sample Questions for Debriefing	
Describe the recent literature concerning protocolized sepsis care. What is the definition of sepsis, severe sepsis and septic shock? Why are the possible reasons why this patient deteriorated into a PEA arrest? What is the surviving sepsis campaign 3 hour bundle? What is Charcot's triad and Reynold's Pentad? What is your approach to vasopressor therapy in sepsis? In post arrest care? What is your approach to ensuring your team understands the urgency of sepsis management? How do you prioritize fluids vs. IV antibiotics? What are some strategies for helping get your team on board?	
Key Moments	
Septic work up	
Persistent hypotension	
PEA and Post Arrest	